

Software Requirements Specification

For
UpTrend

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A handwritten signature in blue ink that reads 'Keshav Sinha'.

Mentor's Signature

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1. INTRODUCTION

1.1 Purpose of the Project

The purpose of the **UpTrend platform** is to provide a structured environment where beginners can learn trading concepts and practice them without financial risk.

- Provide a **risk-free paper trading environment** using virtual currency.
- Help beginners understand **financial markets, trading strategies, and risk management**.
- Bridge the gap between **theoretical knowledge and practical trading experience**.
- Reduce confusion caused by **fragmented online trading resources**.
- Allow users to test trading strategies in a **simulated real-market environment**.
- Provide **performance analytics** to track trading progress.
- Build user confidence before entering **real financial markets**.
- Integrate **educational modules with trading simulation** for better learning.

1.2 Target Beneficiary

The UpTrend platform is designed for individuals who want to learn trading in a structured and practical manner.

- **Beginner traders** who want to understand stock and cryptocurrency markets.
- **Students studying finance, economics, or business** who require practical exposure.
- **Self-learners and professionals** interested in financial literacy and investments.
- Individuals who want to **test trading strategies without risking real money**.
- Users who prefer **structured learning rather than scattered online tutorials**.
- People who want to track and improve their **trading discipline and performance**.
- Educational institutions that require **interactive financial learning tools**.

1.3 Project Scope

The UpTrend project focuses on developing a **web-based trading education and paper trading platform**.

- Development of a **paper trading simulator** that allows users to execute virtual trades.
- Integration of **real-time or delayed market price data using financial APIs**.
- Implementation of a **portfolio management system** to track investments.
- Creation of a **performance analytics dashboard** displaying key trading metrics.
- Development of **structured learning modules for trading education**.
- Building a **responsive web interface** accessible on desktops and mobile devices.
- Implementation of **secure authentication and user management**.
- Ensuring the platform provides a **realistic yet risk-free trading environment**.

2. PROJECT DESCRIPTION

2.1 Reference Algorithm

The system uses several algorithms to simulate trading operations and analyze user performance.

Trading Simulation Algorithm

- Processes buy and sell orders using virtual currency.
- Updates user portfolios based on executed trades.

Portfolio Valuation Algorithm

- Calculates portfolio value based on current market prices.
- Tracks profit and loss for each trade.

Performance Analytics Algorithm

- Computes trading metrics such as win rate and average return.
- Tracks trading consistency and discipline.

Learning Progress Algorithm

- Monitors completion of academy modules.
- Unlocks trading features after basic learning milestones.

2.2 Data/Data Structure

The UpTrend platform manages various types of structured and unstructured data.

Input Data

- User registration information.
- Login credentials and authentication data.
- Trading orders (buy/sell requests).
- Market price data from financial APIs.
- Educational progress data.

Output Data

- Portfolio valuation reports.
- Profit and loss calculations.
- Trading performance metrics.
- Trading history logs.

Data Storage

- MongoDB database used for storing structured data.
- Collections include:
 - Users
 - Trades
 - Portfolio
 - Learning Modules

2.3 SWOT Analysis

The UpTrend platform presents several strengths, weaknesses, opportunities, and threats.

Strengths

- Provides a **risk-free trading environment** for beginners.
- Combines **education and practical trading experience**.
- Uses modern web technologies for scalability.

Weaknesses

- Dependency on **third-party financial APIs**.
- Simulated trading may not replicate real emotional trading behavior.

Opportunities

- Integration of **AI-based trading insights**.
- Development of **social trading communities**.
- Addition of **gamification and leaderboards**.

Threats

- Competition from professional trading simulators.
- API policy changes affecting data availability.

2.4 Project Features

The UpTrend platform includes several core features that enhance the learning and trading experience.

- **Paper Trading Engine**

Simulates buying and selling of assets using virtual money.

- **Trading Dashboard**

Displays capital, win rate, total trades, and analytics.

- **Portfolio Management**

Tracks user holdings and investment value.

- **Academy Learning Module**

Provides lessons on trading basics and strategies.

- **Trading History**

Stores all past transactions.

- **Performance Analytics**

Tracks trading discipline, win rate, and consistency.

2.5 User Classes and Characteristics

The platform supports different categories of users with varying knowledge levels.

Beginner Users

- Limited knowledge of financial markets.
- Require structured learning modules.
- Use paper trading to practice basic strategies.

Intermediate Users

- Have basic understanding of trading.
- Use simulation to test trading strategies.

Advanced Learners

- Focus on improving trading discipline.
- Use analytics to optimize trading performance.

Administrators

- Manage system updates.
- Monitor platform performance and user activity.

2.6 Design and Implementation Constraints

The development of the platform is influenced by several technical constraints.

Technology Constraints

- System built using **MERN Stack**.
- UI designed with **Tailwind CSS**.
- Backend services developed using **Node.js and Express.js**.

Performance Constraints

- Trading engine must process transactions quickly.
- Dashboard must display data updates efficiently.

Security Constraints

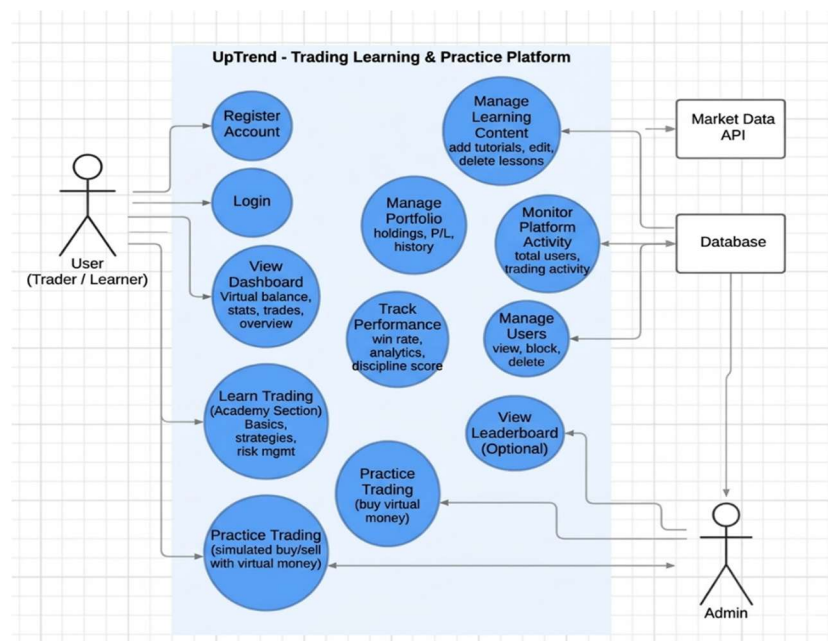
- Secure user authentication using **JWT tokens**.
- Password encryption using **Bcrypt hashing**.

External Dependencies

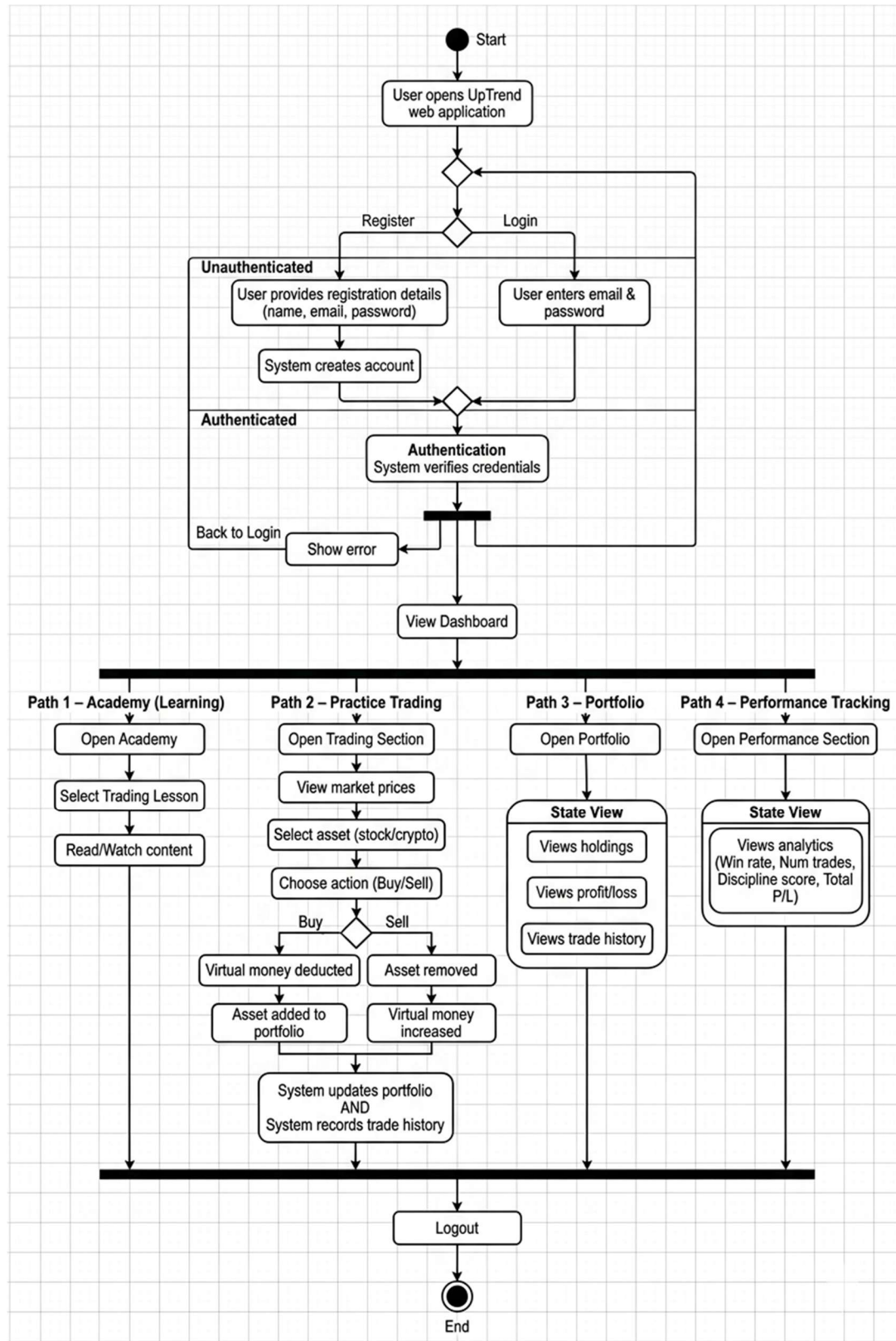
- Financial data APIs for real-time market prices.

2.7 Design Diagrams

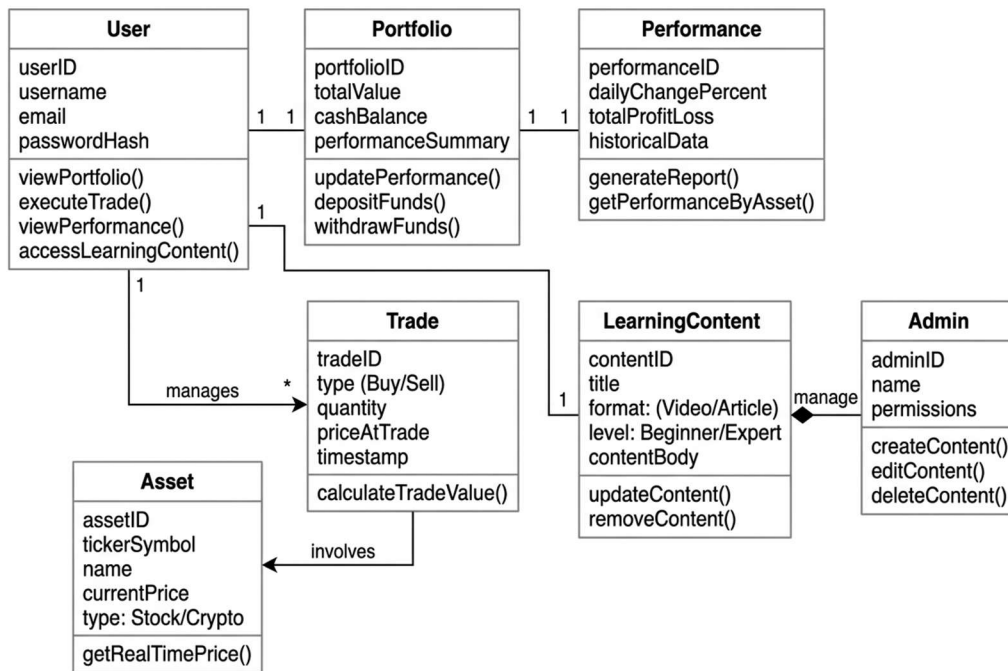
- **Use Case Diagram:**



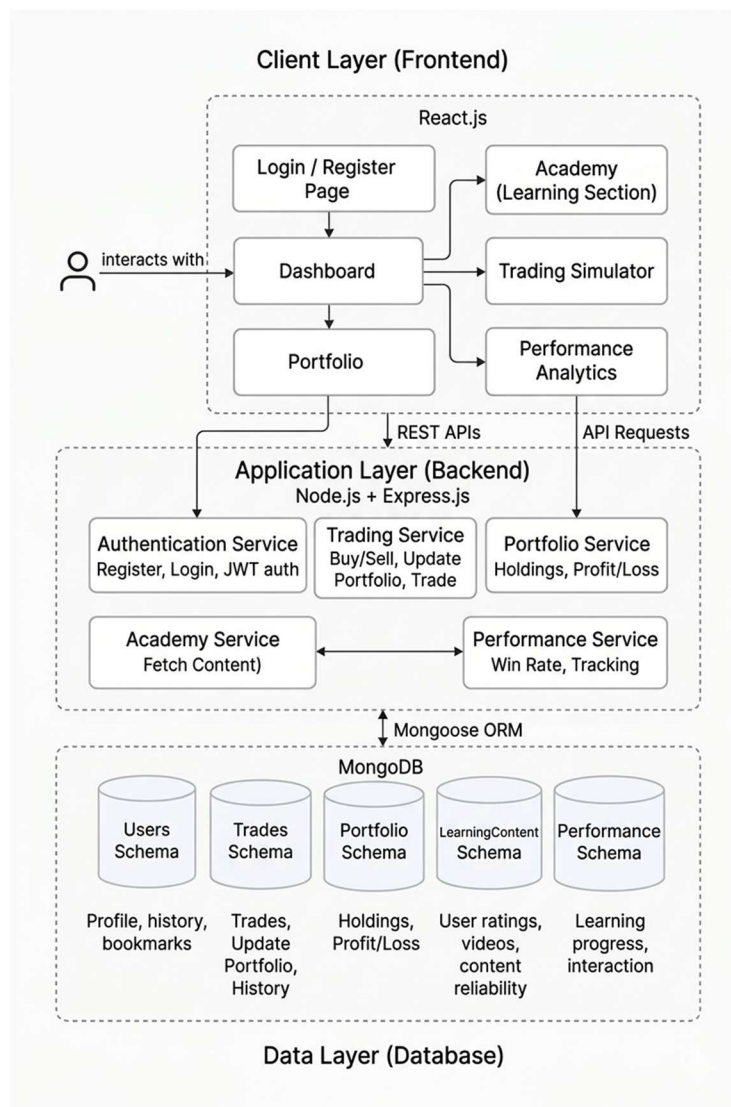
- **Activity Diagram:**



- **Class Diagram:**



- **System Architecture Diagram:**



3. SYSTEM REQUIREMENTS

3.1 User Interface

The UpTrend platform includes several interface components designed for ease of use.

Dashboard

- Displays trading capital and portfolio performance.
- Shows analytics such as win rate and trading discipline.

Trading Page

- Allows users to execute simulated buy and sell orders.
- Displays real-time market prices.

Portfolio Page

- Displays user holdings and investment values.
- Shows profit and loss summaries.

Academy Section

- Provides educational lessons and quizzes.
- Unlocks trading features after completing modules.

Responsive Design

- Accessible on desktops, tablets, and smartphones.

The system integrates with several external software tools.

- Financial APIs such as **Alpha Vantage** and **Yahoo Finance**.
- Charting libraries like **Recharts** or **Chart.js**.
- Backend APIs developed using **Node.js** and **Express.js**.
- Frontend developed using **React.js**.
- Version control managed using **Git** and **GitHub**.

These integrations ensure real-time data updates and smooth communication between system components.

3.1 Database Interface

The platform uses **MongoDB** as the primary database.

- Stores user profiles and authentication data.
- Maintains records of all simulated trades.
- Tracks portfolio holdings and profit/loss history.
- Saves user progress in academy modules.

Database Collections

- Users
- Trades
- Portfolios
- Learning Modules

This database design allows efficient storage and retrieval of large volumes of trading and user activity data.

3.2 Protocols

The system relies on several communication protocols.

- **HTTP/HTTPS Protocol**
 - Used for communication between frontend and backend.
- **REST APIs**
 - Handle requests such as user login and trade execution.
- **WebSockets**
 - Enable real-time updates for market prices.
- **Secure Authentication**
 - Uses encrypted tokens for session management.

These protocols ensure secure and reliable communication between the application components.

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements

- Summarization and ranking must complete in under 5 seconds per query.
- 3D visualization rendering should handle up to 50 nodes without lag on mid-range devices.
- Offline access retrieves cached data instantly.

4.2 Security Requirements

- User data encrypted in transit and at rest.
- No storage of copyrighted source content; summaries generated on the-fly.
- Authentication for personalized features.

4.3 Software Quality Attributes

- **Usability:** Intuitive navigation and responsive design.
- **Reliability:** Handles diverse inputs without crashes; 99% uptime.
- **Adaptability:** Supports multiple devices and browsers.

5. OTHER REQUIREMENTS

OTHER REQUIREMENTS

Future improvements may include:

- AI trading insights
- Leaderboards and gamification
- Social trading features
- Advanced order types (limit orders, stop-loss)

Appendix A: Glossary

Paper Trading – Simulated trading using virtual money.

P&L – Profit and Loss calculation.

MERN Stack – MongoDB, Express.js, React.js, Node.js.

API – Application Programming Interface used for data fetching.

Appendix B: Analysis Model

Includes:

- Entity Relationship Model
- System Architecture
- Data Flow between frontend and backend

Appendix C: Issues List

To be filled during testing and development stages.